

Poisson Matching Lemma and Its Applications

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Abstract—This project concerns Poisson Matching Lemma (PML) [1], Strong Functional Representation Lemma (SFRL) [2], and their applications to one-shot achievability problems in Information Theory. For given random variables X and Y , functional representation means finding a random variable Z , and a deterministic function g such that Z is independent of X and Y can be represented as $g(X, Z)$. We can think of the SFRL as a tool that provides functional representation with an Information Theoretic bound on the conditional entropy of Y given Z (i.e. $H(Y|Z) < I(X, Y) + \log(I(X, Y) + 1) + 4$). In this report, first, we present a technique that we call Universal SFRL (USFRL) which is the universal version of the SFRL and its application to the One-Shot Universal Channel Simulation problem. Then, we examine our attempt at using the USFRL for the Universal Source Coding Problem by comparing it to how the SFRL is used for the Rate-Distortion problem in [2]. Finally, we present the PML [1] which is an extended version of the work done in [2]. In this report, we summarize these papers, present our related works, and provide potential research directions.

Index Terms—Strong Functional Representation Lemma, Poisson Matching Lemma, One-Shot, Achievability, Channel Simulation, Rate Distortion, Universal Compression, Channel Coding

I. INTRODUCTION

STRONG functional representation lemma (SFRL), which is introduced in [2] with its applications to some fundamental coding problems in Information Theory, is a technique that provides a functional representation for a given random variable Y in terms of a given random variable X , and a random variable of our choice Z which is independent of X . Several other methods provides functional representation other than the SFRL, but what makes it unique is that it helps us to make this functional representation in a way that Z is most informative about Y (i.e. minimizes the conditional entropy $H(Y|Z)$). In section II of [3], we have presented some functional representation methods with examples and natural limits for functional representation. In section II of this report, we will present the SFRL which is the method that we use for making Z as informative as possible about Y in the functional representation problem. Then, using the same Poisson Functional Representation trick as we used in the proof of the SFRL, we will introduce and prove the Universal Strong Functional Representation Lemma (USFRL) in section III and present its application to One-Shot Universal Channel Simulation problem in section IV.

On top of the applications of the SFRL given in [2] and the One-Shot Universal Channel Simulation of the USFRL, we will consider the Universal Compression Problem. It is known that there are several Universal Compression algorithms for which the redundancy diminishes (i.e. the number of bits sent per symbol approaches to the entropy of the source) in the

asymptotical case. However, it is not possible to have zero redundancy for one shot. To find a one-shot scheme with small redundancy for Universal Compression, we will first present the Lossy Source Coding application of the SFRL in [2] in V, and then we will present our approach to Universal Compression problem in section VI by comparing it to the Lossy Source Coding application of the SFRL in section V.

Then, we will continue with the Poisson Matching Lemma (PML) and its applications to some one-shot achievability problems in Information Theory, which is introduced in [1], extends the work done in [2] by providing an upper bound for the probability of a random variable $\tilde{U}_Q$ given by the probability measure of Q being different than a *known* random variable $\tilde{U}_P$ given by the probability measure of P in terms of the density of these probability measures P and Q at the value of the random variable $\tilde{U}_P$ which is $\Pr\{\tilde{U}_Q \neq \tilde{U}_P | \tilde{U}_P\} \leq 1 - (1 + \frac{dP}{dQ}(\tilde{U}_P))^{-1}$. We are going to explain the PML with its proof and present a variation of it called the Conditional Poisson Matching Lemma (CPML) in the section VII. Then, we are going to present an application of the CMPL which is One-Shot Channel Coding in section VIII.

Last but not least, I will talk about the goals of this project and what work I am planning to do to continue this project. I will mention the other applications of the PML given in [1] and I will discuss other potential applications of them in the section IX.

Note that we used "log" as it is in base 2 throughout this report.

II. STRONG FUNCTIONAL REPRESENTATION LEMMA

Here is the statement of the SFRL which is the main theorem of [2].

Theorem 1 (Strong Functional Representation Lemma):
Given random variables X and Y , we can find a random variable Z and a function g such that:

$$-X \perp Z$$

$$-Y = g(X, Z)$$

$$-H(Y|Z) \leq I(X, Y) + \log(I(X, Y) + 1) + 4$$

Thus, the SFRL provides a very good bound of order $O(n + \log(n))$ for the natural limit $n = I(X, Y) \leq H(Y|Z)$.

Now, let's examine the construction for Z and g suggested in [2]. Z is given as the Poisson point process as follows:

$$Z = \{(T_i, \tilde{Y}_i)\}_{i=1}^{\inf}$$

where T_i 's are the event times of a Poisson process with rate 1 and $\tilde{Y}_i$'s are i.i.d random variables distributed as Y . Then, Z is clearly independent of X by its definition.

Moreover, we can choose $g(X, Z)$ as

$$g(x, z) = g(x, \{\tilde{y}_i, t_i\}) = \tilde{y}_{k(x, \{\tilde{y}_i, t_i\})}$$

where

$$k(x, \{\tilde{y}_i, t_i\}) = \arg \min_i t_i \frac{dP_Y}{dP_{Y|X=x}}(\tilde{y}_i)$$

In [3], we explained why this scheme works as a functional representation for the discrete case. Now, let's prove the information theoretical bound on $H(Y|Z)$:

Proof: (for the information theoretical bound)

We have $Z = \{(T_i, \tilde{Y}_i)\}_{i \in N}$ where $T_1, T_2 - T_1, T_3 - T_2, \dots$ are i.i.d. with $\sim \exp(1)$ and $\tilde{Y}_i$'s are i.i.d. with $\sim p_Y$. From Z , we want to deterministically produce a $\tilde{Y}$ with any given distribution q_Y .

Later, we will select this distribution q_Y as $p_{Y|X}$. Therefore, $(X, \tilde{Y}) \sim P_{XY}$ which means $\tilde{Y} = Y$.

Let $\tilde{T}_i := T_i \frac{p_Y(\tilde{Y}_i)}{q_Y(\tilde{Y}_i)} \forall i \in N$ and let $K_1 = \arg \min_i \tilde{T}_i$, $K_2 = \arg \min_i \tilde{T}_{i \neq K_1}$, $K_3 = \arg \min_i \tilde{T}_{i \neq K_1 \& i \neq K_2}$. Then, $\{(\tilde{T}_{K_i}, \tilde{Y}_{K_i})\}_{i \in N}$ will be a Poisson point process with intensity measure q_Y .

Conditioned on $Y_k = y$, $\tilde{T}_{K_1} = t$; K can be found as 1 + the number of points t_0 such that:

$$t_0 \frac{q_Y(\tilde{y})}{p_Y(\tilde{y})} < t \frac{q_Y(y)}{p_Y(y)}$$

Moreover, t_0 must be greater than $\tilde{T}_{K_1} = t$. So we have $t < t_0 < t \frac{q_Y(y)}{p_Y(y)} \frac{p_Y(\tilde{y})}{q_Y(\tilde{y})}$. Thus:

$$K = 1 + \sum_{\tilde{y} \neq y} \text{Poisson}\left(\left[t \frac{q_Y(y)}{p_Y(y)} \frac{p_Y(\tilde{y})}{q_Y(\tilde{y})} - t\right]^+ q_Y(\tilde{y})\right)$$

Taking the expectation of both sides:

$$\mathbb{E}[K | \tilde{Y}_k = y, \tilde{T}_{K_1} = t] = 1 + \sum_{\tilde{y} \neq y} \left[t \frac{q_Y(y)}{p_Y(y)} \frac{p_Y(\tilde{y})}{q_Y(\tilde{y})} - t\right]^+ q_Y(\tilde{y})$$

$$\implies \mathbb{E}[\log K | \tilde{Y}_k = y, \tilde{T}_{K_1} = t] \leq \log(\mathbb{E}[K | \tilde{Y}_k = y, \tilde{T}_{K_1} = t])$$

$$\leq \log\left(1 + \sum_{\tilde{y} \neq y} \left[t \frac{q_Y(y)}{p_Y(y)} \frac{p_Y(\tilde{y})}{q_Y(\tilde{y})} - t\right]^+ q_Y(\tilde{y})\right)$$

where the first inequality follows from Jensen's Inequality. Then, averaging over $\tilde{Y}_k$ and $\tilde{T}_{K_1}$:

$$\mathbb{E}[\log K] =$$

$$\sum_y q_Y(y) \int_0^\infty e^{-t} \log\left(1 + \sum_{\tilde{y} \neq y} \left[t \frac{q_Y(y)}{p_Y(y)} \frac{p_Y(\tilde{y})}{q_Y(\tilde{y})} - t\right]^+ q_Y(\tilde{y})\right) dt \leq (1)$$

$$\sum_y q_Y(y) \log\left(\int_0^\infty e^{-t} \left(1 + \sum_{\tilde{y} \neq y} \left[t \frac{q_Y(y)}{p_Y(y)} \frac{p_Y(\tilde{y})}{q_Y(\tilde{y})} - t\right]^+ q_Y(\tilde{y})\right) dt\right)$$

$$\begin{aligned} &= \sum_y q_Y(y) \log\left(1 + \sum_{\tilde{y} \neq y} \left[t \frac{q_Y(y)}{p_Y(y)} \frac{p_Y(\tilde{y})}{q_Y(\tilde{y})} - t\right]^+ q_Y(\tilde{y})\right) \\ &\leq \sum_y q_Y(y) \log\left(1 + \frac{q_Y(y)}{p_Y(y)}\right) \leq (2) D(q||p) + 1 \end{aligned}$$

where (1) and (2) come from Jensen Inequality. Then, using an inequality on non-negative integer-valued random variables, we get:

$$H(K) \leq E(\log K) + \log(E[\log K] + 1) + 1$$

$$\leq D(q||p) + \log(D(q||p) + 1) + c$$

for some constant $c < 4$. Then, since Y is a deterministic function of K and Z :

$$H(Y|Z) \leq H(K|Z) \leq H(K) \leq D(q||p) + \log(D(q||p) + 1) + c$$

This holds for a given $X = x$. Choosing the distributions as $q = p_{Y|X=x}$ and $p = p_Y$ and taking the average over X , we get:

$$H(Y|Z) \leq I(X, Y) + \log(I(X, Y) + 1) + c$$

■

As we can see, this proof works not only when $q_Y = p_{Y|X}$, but for any distribution q_Y . Therefore, this is a more general way to prove the SFRL than the one given in [2]. We will use the ideas in this proof to show the USFRL in the section III.

III. UNIVERSAL STRONG FUNCTIONAL REPRESENTATION LEMMA

Theorem 2 (Universal Strong Functional Representation Lemma):

Given a distribution $p_{Y|X}$, we can find a distribution p_Z and a function $g : (x, z) \rightarrow y$ such that:

$$-(x, f(x, z)) \sim p_{XY}, \forall p_X$$

$$-X \perp Z$$

$$-H(Y|Z) \leq C(X \rightarrow Y) + \log(C(X \rightarrow Y) + 1) + c$$

where Z is a random variable with distribution p_Z and c is some constant.

Proof: (for the case where $\mathcal{X}$ has finite cardinality)

Given $p_{Y|1}, p_{Y|2}, \dots, p_{Y||\mathcal{X}|}$, let's pick any distribution p on $\mathcal{Y}$. Then, let's generate a random variable Z with respect to p as $Z = \{(T_i, Y_i)\}_{i \in N}$ where $T_1, T_2 - T_1, T_3 - T_2, \dots$ are i.i.d. with $\sim \exp(1)$ and Y_i 's are i.i.d. with $\sim p$. Then, Z will be independent of X .

Now, consider the following method to generate $f(x, z)$ from a given x . Given x and $z = \{(t_i, y_i)\}_{i \in N}$ (a realization of Z), let $q(y) = p_{Y|X}(y|x)$. Then, we use the same procedure as in the proof of Theorem 1 to find $k =$

$k(z, q) = \arg \min_i t_i \frac{p(y_i)}{q(y_i)}$ and output $f(x, z) = y_k$. This way, $(x, y_k = f(x, z)) \sim p_{XY}$ as we showed in the proof of Theorem 1.

Moreover, given $X = x$, we acquired k with the same procedure as in the proof of Theorem 1. Thus, the following inequality holds:

$$E[\log K|X = x] \leq D(q||p) + c_1 = D(p_{Y|X}(\cdot|X = x)||p) + c_1$$

for some constant c_1 . Then, averaging over X

$$\begin{aligned} E[\log K] &= \sum_x E[\log K|X = x] \Pr[\{X = x\}] \\ &\leq \left(\sum_x p_X(x) D(p_{Y|X=x}(\cdot|X)||p) \right) + c_1 \end{aligned}$$

Moreover, since K is a non-negative integer-valued random variable, we have:

$$H(K) \leq E[\log K] + \log(E[\log K + 1]) + 1 \leq$$

$$G\left(\sum_x p_X(x) D(p_{Y|X=x}(\cdot|X)||p)\right)$$

where $G(D) = D + c_1 + \log(D + c_1) + 1$. Consequently,

$$\begin{aligned} H(Y|Z) &\stackrel{(1)}{\leq} H(K|Z) \leq H(K) \leq \\ &G\left(\sum_x p_X(x) D(p_{Y|X=x}(\cdot|X)||p)\right) \end{aligned}$$

where (1) is because Y is a deterministic function of K and Z .

By the properties of mutual information, we know $\min_p \max_{x \in \mathcal{X}} D(p_{Y|X=x}(\cdot|X)||p) = C(X \rightarrow Y)$. Assume that this minimum is achieved by p^* . Recall that we picked p in a way that it could be any distribution on $\mathcal{Y}$ at the beginning of the proof. Then, choosing p as p^* , we acquire:

$$\begin{aligned} H(Y|Z) &\leq G(C(X \rightarrow Y)) \leq \\ &C(X \rightarrow Y) + \log(C(X \rightarrow Y) + 1) + c_2 \end{aligned}$$

■

IV. ONE-SHOT UNIVERSAL CHANNEL SIMULATION

One useful application of the USFRL is the One-Shot Universal Channel Simulation. We discussed the One-Shot Channel Simulation problem in [3] using the SFRL. The difference is the source probability (p_X) is not known in the universal case. The framework for the problem is provided in Fig. 1 below.

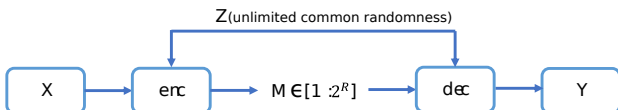


Figure 1. One-Shot Universal Channel Simulation Scheme

We want to create a conditional distribution $P_{Y|X}$ by designing an encoder and decoder functions that are allowed to share unlimited common randomness. Here is the exact statement for the problem:

Definition 1 (One-Shot Channel Simulation Problem): For given random variables X with an unknown distribution p_X , we want to simulate a channel $P_{Y|X}$ by designing an encoder, decoder pair (enc, dec) with unlimited common randomness Z such that we can represent the message M (which is the output of the encoder and the input of the decoder) in as small number of bits as possible. If we can do that in R bits, then R is called an achievable rate.

It turns out that an achievable rate R should be at least the capacity of the channel $C(P_{Y|X}) = \max_{P_X} I(X; Y)$ by the theorem given by Bennett, Shor, Smolin, and Thapliyal in [4] stated below:

Theorem 3 (BSST Theorem, 2002): Suppose that $\inf\{R : R \text{ is an achievable rate for the channel simulation problem}\}$ is called as $R_{\text{channel simulation}}^*$. Then:

$$R_{\text{channel simulation}}^* = \max_{P_X} I(X; Y)$$

Now, let's see how we can approach this problem by utilizing the USFRL.

We want to simulate a channel $P_{Y|X}$ for a given input X with an unknown distribution p_X . Then, by the USFRL, $\exists$ a random variable Z and a function $f : x, z \rightarrow y$ such that $\forall p_X$ the distribution of $(x, f(x, z))$ is given by $p_{XY} = p_X p_{Y|X}$, and $H(Y|Z) \leq C(X \rightarrow Y) + \log(C(X \rightarrow Y) + 1) + c$ for some constant c . We will use this Z as the common randomness between the encoder and decoder.

Here, encoder assumes any distribution p'_X on $\mathcal{X}$ and acquire the corresponding distribution on $\mathcal{Y}$ with respect to given $p_{Y|X}$ as $p_Y = \sum_x p'_X(x) p_{Y|X}(\cdot|x)$. Then, $f(X, Z) \sim p_Y$ by the USFRL. Then, since Z is common between encoder and decoder $f(x, z)$ can be sent to the decoder by less than $H(Y|Z) + 1$ bits with the Huffman coding. And, the decoder recovers $f(x, z) \sim p_Y$.

Last but not least, we know that $H(Y|Z) + 1 \leq C(X \rightarrow Y) + \log(C(X \rightarrow Y) + 1) + c_2$ for some constant c_2 for all p_X (including p'_X) by the USFRL. Thus, we are only a logarithmic term apart from achieving the minimum achievable rate while using a one-shot scheme.

V. LOSSY SOURCE CODING

In this part, let us revisit the Lossy Source Coding problem that we discussed in [3]. We will point out its differences from the Universal Compression Problem while we are examining our attempt to apply the USFRL to Universal Compression. Here is the exact statement for the Lossy Source Coding Problem:

Definition 2 (Lossy Source Coding Problem): For a given random variable X , we want to design an encoder, decoder pair (enc, dec) such that $E[d(X, Y)] \leq D$ where Y is the random variable that represents the output of the decoder. We want to design this (enc, dec) such that we represent the message M (which is the output of the encoder and the input of the decoder) in as small a number of bits as possible. If

we can do that in R bits, then R is called an achievable rate. Moreover, for a given random variable X and a fixed distortion function d , we define $R(D) = \inf\{R : R \text{ is achievable}\}$. The framework is given in Fig. 2.

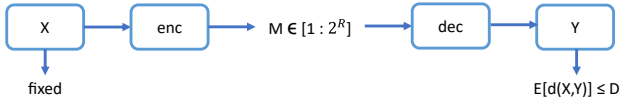


Figure 2. Lossy Source Coding Scheme

The natural lower bound for the achievable rate of the Lossy Source Coding problem is given by the Lossy Source Coding Theorem as follows:

$$R(D) = \min_{P_{Y|X}: E[d(X, Y)] \leq D} I(X; Y)$$

Again by the Lossy Source Coding Theorem, this limit is achievable. However, the typicality ([enc, dec]) is used in the proof and it is not implementable as it requires to construct and store an $n \times 2^{nR(D)}$ dimensional random matrix as the codebook.

The SFRL can be used to design an ([enc, dec]) pair which is easy to implement and achieves a rate of $\hat{R}$ for all $\hat{R} \geq R(D) + \log(R(D) + 1) + c$ for some constant c as shown in [2]. Therefore, we can use the SFRL to design a Lossy Source Coding Scheme with a reasonable rate (in the order of $O(R(D) + \log(R(D)))$) for the natural lower limit $R(D)$. Moreover, the design given in [2] is almost deterministic, so it does not use a random encoder, unlike the typicality encoder that achieves $R(D)$. The reason we call this design almost deterministic is that it actually uses only 1-bit of randomness for codebook generation in the proof. Now, let's state this as a theorem and present its proof.

Theorem 4: For the Lossy Source Coding Problem given in definition 2, $\exists$ an ([enc, dec]) pair such that $R(D) + \log(R(D) + 1) + c$ is an achievable rate in one-shot.

Proof: By the Lossy Source Coding Theorem, we know that $R(D) = \min_{P_{Y|X}: E[d(X, Y)] \leq D} I(X; Y)$. Then, $\exists p_{Y^*|X}^*$ such that $I(X; Y^*) = R(D)$ where Y^* is the output of the channel $p_{Y^*|X}^*$ with input X . Moreover, $E[d(X; Y^*)] \leq D$.

Now, using (X, Y^*) , we will acquire a random variable $Z = \{(T_i, Y_i^*)\}_{i \in N}$ and a function g such that $X \perp Z$, $g(X, Z) = Y^*$, and $H(Y^*|Z) \leq I(X; Y^*) + \log(I(X; Y^*) + 1) + c = R(D) + \log(R(D) + 1) + c$.

Now, consider the following set:

$$A = \{(\underbrace{H(g(X, z)), E[d(X, g(X, z))]}_{:=a(z)}) : z \in \mathcal{Z}\}$$

$$\begin{aligned} \Rightarrow E[a(z)] &= [E[H(g(X, z))], E[E[d(X, g(X, z))]] = \\ &= [E[H(Y^*|Z = z)], E[d(X, Y^*)]] = [H(Y^*|Z), E[d(X, Y^*)]] \\ &\leq [R(D) + \log(R(D) + 1) + c, D] \end{aligned}$$

We know that $E[a(z)] \in \text{ConvexHull}\{A\}$. Then, by Caratheodory's Theorem, there exist 2 points (z_0, z_1) in A and a number $\lambda \in [0, 1]$ such that:

$$\lambda a(z_0) + (1 - \lambda)a(z_1) \preceq E[a(z)]$$

$$\preceq [R(D) + \log(R(D) + 1) + c, D]$$

Then, choosing $\tilde{Y} = g(X, z_Q)$ where $Q \sim \text{Bernoulli}(1 - \lambda)$, we get:

$$E[d(X, \tilde{Y})] = E[d(X, g(X, z_Q))] \leq [E[a(z)]]_2 \leq D$$

And, we also get:

$$H(\tilde{Y}) \leq H(Y, Q) = H(\tilde{Y}|Q) + \underbrace{H(Q)}_{\leq 1} \leq$$

$$[\lambda a(z_0) + (1 - \lambda)a(z_1)]_1 + 1 \leq [E[a(z)]]_1 + 1$$

$$\leq R(D) + \log(R(D) + 1) + \underbrace{c'}_{c+1}$$

Then, we can tell $\tilde{Y}$ by using less than $H(\tilde{Y}) + 1$ bits in average. Note that the encoder does all of these operations and sends $\tilde{Y}$ using the Huffman code, and the decoder just decodes this coded message. ■

VI. UNIVERSAL COMPRESSION

One of the biggest problems that we considered in this project was using the SFRL for the Universal Compression problem. This is actually how we end up with the USFRL. In this section, I will summarize the main approaches that we tried to apply the SFRL to One-Shot Universal Compression with Common Randomness. Here is how we construct the problem:

Definition 3 (One-Shot Universal Compression Problem):

For a given random variable $Y \sim p_Y$, we know that $p_Y \in \{p_{Y|X=1}, p_{Y|X=2}, \dots, p_{Y|X=M}\}$ and $X \in \{1, 2, \dots, M\}$ with an unknown distribution p_X .

Note that if we assume p_X is both known by decoder and encoder, then the problem would be equivalent to the classical source coding problem.

Although our first intention in creating the USFRL was to apply it to the Universal Compression, we can not apply it to this problem in a straightforward way. The reason behind this is that the USFRL provides us with a random variable with the demanded distribution, but it is not useful for providing the exact same realization that we need for One-shot Universal Compression. To see this in a better way, let's look closer at one of the failed attempts of applying the USFRL to the One-shot Universal Compression Problem:

Assume on top of $p_{Y|X}$, we also know $p_{X|Y}$. Then, by the USFRL, $\exists Z \sim p_Z$ and $f : y, z \rightarrow x$ such that $(y, f(y, z)) \sim p_{YX}$ and $H(X|Z) \leq C(X \rightarrow Y) + \log(C(X \rightarrow Y) + 1) + c$ hold $\forall p_Y$. We will use this Z as the common randomness between the encoder and decoder. (Fig. 3)

Then, given $Y = y$ and $Z = z$, the encoder produces $X' = x'$ with distribution $p_{X|Y=y}$. Here, X' has the same

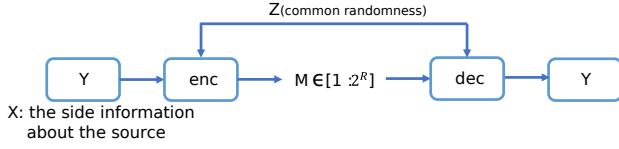


Figure 3. One-Shot Universal Compression Scheme

distribution X , but their realizations, x and x' , are not necessarily the same. If x and x' were equal, we could claim that x could be sent to the decoder by less than $H(X|Z) + 1$ bits in expectation using Huffman coding. Moreover, once x also becomes available to the decoder, the encoder could also tell the message y by almost $H(Y|X)$ bits in expectation using Huffman coding. However, all of these do not work because the USFRL only helps us to come up with a random variable with the demanded distribution (i.e. $X' \sim X$), but it does not guarantee that they have the same realizations (i.e. $x' = x$ not guaranteed).

To understand this issue even further, it might be useful to recall how the SFRL applied to the Lossy Compression problem in section V. We used the SFRL to obtain a set of random variables with the demanded distributions (i.e. the set that we called A) and, then, we used the fact that there exists a random variable in this set that satisfies the distortion constraint.

However, as it was in Lossy Compression, there might be other ways in which we use some set of distributions to use the USFRL for the Universal Compression problem. Therefore, this is going to be one of the future directions of this project.

VII. POISSON MATCHING LEMMA

Now, let's present the Poisson Matching Lemma (PML) given in [1]. It is built on top of the work in [2] and its idea is based on the properties of Poisson Point Processes just like the SFRL.

Now, let's introduce the PML which is the main theorem of [1]:

Theorem 5 (Poisson Matching Lemma): Let $\{\bar{U}_i, T_i\}_{i \in N}$ be the points of a Poisson process with intensity measure μ (Lebesgue Measure). Also, let P and Q be some probability measures on $\mathcal{U}$ with $P, Q \ll \mu$ (i.e. absolute continuity). Then, $\tilde{U}_P$ and $\tilde{U}_Q$ are defined as:

$$\tilde{U}_P(\{\bar{U}_i, T_i\}_{i \in N}) := \bar{U}_{K_P(\{\bar{U}_i, T_i\}_{i \in N})}$$

$$\tilde{U}_Q(\{\bar{U}_i, T_i\}_{i \in N}) := \bar{U}_{K_Q(\{\bar{U}_i, T_i\}_{i \in N})}$$

where

$$K_P(\{\bar{U}_i, T_i\}_{i \in N}) := \operatorname{argmin}_{i: \frac{dP}{d\mu}(\bar{U}_i) > 0} T_i \frac{dP}{d\mu}(\bar{U}_i)^{-1}$$

$$K_Q(\{\bar{U}_i, T_i\}_{i \in N}) := \operatorname{argmin}_{i: \frac{dQ}{d\mu}(\bar{U}_i) > 0} T_i \frac{dQ}{d\mu}(\bar{U}_i)^{-1}$$

with an arbitrary tie-breaking that happens with zero probability. Moreover, for the sake of easier notation, we will notate

$\tilde{U}_P(\{\bar{U}_i, T_i\}_{i \in N})$ as simply $\tilde{U}_P$ if $\{\bar{U}_i, T_i\}_{i \in N}$ is clear from the context.

Then, we have the following inequality almost surely:

$$\Pr(\{\tilde{U}_Q \neq \tilde{U}_P | \tilde{U}_P = u\}) \leq 1 - \frac{1}{1 + \frac{dP}{dQ}(\tilde{U}_P)}$$

Proof: (for discrete alphabets $\mathcal{U}$)

Let $Z_u \sim \text{Exp}(1)$ be i.i.d random variables. Then, we can represent $\tilde{U}_P$ as $\operatorname{argmin}_u Z_u / P(u)$ by the properties of Poisson processes when we think Z_u 's as the event times for a Poisson process with rate 1. Let $\tilde{Z}_P := \frac{Z_{\tilde{U}_P}}{P(\tilde{U}_P)}$.

Conditional on $\tilde{U}_P = u$ and $\tilde{Z}_P = \tilde{z}$, we know $\frac{Z_v}{P(v)} \tilde{z}$ for all $v \in \mathcal{U}$ different than u . Then, by the memoryless property of exponential distribution, we have $Z_v - P(v) \tilde{z} \sim \text{Exp}(1)$ because $P(Z_v > c) = P(Z_v - P(v) \tilde{z} > c | Z_v P(v) \tilde{z})$ for any arbitrary c due to memoryless property and Z_v is known to be exponentially distributed with rate 1. Moreover, $Z_v - P(v) \tilde{z}$'s are independent across v 's and independent of $\tilde{Z}_P$. The first independency claim follows due to the independency of Z_v 's across v 's. Here, v is any element in $\mathcal{U}$ different than u , therefore for $u \in \mathcal{U}$, we see $Z_v - P(v) \tilde{z}$'s are independent of $\tilde{Z}_P$ due to the independency of Z_v 's with Z_u .

Now, using all these observations, let's prove the inequality given in the statement of the Poisson Matching Lemma:

$$\Pr(\{\tilde{U}_Q \neq \tilde{U}_P | \tilde{U}_P = u\}) = \Pr(\{\min_{v \neq u} \frac{Z_v}{Q(v)} \neq \frac{Z_u}{Q(u)} | \tilde{U}_P = u\})$$

$$\leq \Pr(\{\min_{v \neq u} \frac{Z_v - P(v) \tilde{Z}_P}{Q(v)} \neq \frac{\tilde{Z}_P P(u)}{Q(u)} | \tilde{U}_P = u\})$$

$$= \Pr(\{\text{Exp}(1 - Q(u)) \leq \text{Exp}(\frac{Q(u)}{P(u)}) | \tilde{U}_P = u\})$$

for independent exponential distributions. The last line is because $Z_v - P(v) \tilde{Z}_P$'s are independent exponential distributions with rate 1, thus $\frac{Z_v - P(v) \tilde{Z}_P}{Q(v)}$'s are independent exponential distributions with rate $Q(v)$'s, then $\min_{v \neq u} \frac{Z_v - P(v) \tilde{Z}_P}{Q(v)}$ is an exponential distribution with rate $\sum_{v \neq u} Q(v) = 1 - Q(u)$. This is because finding the minimum over $\frac{Z_v - P(v) \tilde{Z}_P}{Q(v)}$'s is like finding the first event time of the combination of some Poisson processes with rates $Q(v)$'s. Thus, the combination becomes another Poisson process with rate $\sum_{v \neq u} Q(v)$ and the first event time of this Poisson process (which is $\min_{v \neq u} \frac{Z_v - P(v) \tilde{Z}_P}{Q(v)}$) is an exponential distribution with rate $\sum_{v \neq u} Q(v)$.

Then, we have the following inequality:

$$\Pr(\{\tilde{U}_Q \neq \tilde{U}_P | \tilde{U}_P = u\})$$

$$\leq \Pr(\{\text{Exp}(1 - Q(u)) \leq \text{Exp}(\frac{Q(u)}{P(u)}) | \tilde{U}_P = u\})$$

Here, we can get rid of the condition for the right-hand side of the inequality because we already reduced the problem to comparing independent exponential distributions. Then:

$$\begin{aligned} \Pr(\{\tilde{U}_Q \neq \tilde{U}_P | \tilde{U}_P = u\}) &\leq \Pr(\{\text{Exp}(1-Q(u)) \leq \text{Exp}(\frac{Q(u)}{P(u)})\}) \\ &= \frac{1-Q(u)}{1-Q(u) + \frac{Q(u)}{P(u)}} \leq \frac{1}{1 + \frac{Q(u)}{P(u)}} = 1 - \frac{1}{1 + \frac{P(u)}{Q(u)}} \end{aligned}$$

Note that the PML can be applied to a lot of problems in Information Theory such as various settings of Channel Simulation and the Lossy Source Coding Problems. These applications are listed as channels with state information at the encoder, Lossy Source Coding with side information at the decoder, joint source-channel coding, broadcast channels, distributed Lossy Source Coding, multiple access channels, and channel resolvability in [1]. To apply the PML to these problems, we might find helpful to use a variation of the PML called Conditional Poisson Matching Lemma (CPML). Let's state it here (in finite alphabet case) as a theorem without giving its proof. Its proof is given in [1].

Theorem 6 (Conditional Poisson Matching Lemma): Fix a distribution p_{XUY} and a probability kernel $Q_{U|Y}$. Generate X according to p_X . Then, let $\{(\tilde{U}_i, T_i)\}_{i \in \mathbb{N}}$ be the points of a Poisson point process. We generate U according to $p_{U|X}$ as $U = \tilde{U}_{p_{U|X}(\cdot|X)}$ where we defined the relation between $\tilde{U}$ and $\tilde{U}$ in the same way as in Theorem 5. Then, by Poisson matching, the distribution of $U = \tilde{U}_{p_{U|X}(\cdot|X)}$ will be $p_{U|X}$. And, finally we generate Y according to $p_{Y|XU}$ such that $Y \rightarrow (X, U) \rightarrow \{(\tilde{U}_i, T_i)\}_{i \in \mathbb{N}}$ is a Markov chain. Then:

$$\Pr(\{\tilde{U}_{Q_{U|Y}(\cdot|Y)} \neq U | X, U, Y\}) \leq 1 - \frac{1}{1 + \frac{P_{U|X}(\cdot|X)}{Q_{U|Y}(\cdot|Y)}(U)}$$

It is worth pointing out that we call CMPL given on the quaternary of $(X, U, Y, Q_{U|Y})$ for this theorem.

VIII. ONE-SHOT CHANNEL CODING

In this section, we will look at one of the most important applications of the PML. Here is the problem setup:

Definition 4 (One-Shot Channel Coding Problem):

Assume we have a message described by a random variable M with a uniform distribution $\text{Unif}[1 : L]$ that we want to send through a channel described by the transition probability $p_{Y|X}$. Upon observing M , the encoder produces X which is sent through the channel $p_{Y|X}$. Then, the decoder observes Y , and recovers $\hat{M}$ with error probability $P_e = \Pr(\{M \neq \hat{M}\})$. We want to design a pair of encoder and decoder functions such that P_e is small.

For the One-shot Channel Coding Problem, we can refer to the scheme in Fig. 4

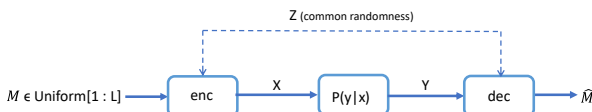


Figure 4. One-shot Channel Coding Scheme

Note that we represented the common randomness between encoder and decoder with a dotted line in Fig. 4. The reason is that [1] proves that the bound in the proof of Theorem 7 is achievable without the use of common randomness although we used the common randomness in the proof of Theorem 7. In this report, for the sake of simplicity, we will only see the proof for the case where we used common randomness. For this case, [1] proposes an upper bound for the error probability in the following theorem:

Theorem 7: Fix any distribution P_X on X . Then, there exists a pair of encoder and decoder such that:

$$P_e \leq E[1 - \frac{1}{1 + L2^{-i(X;Y)}}]$$

$$\text{where } i(X;Y) := \log(\frac{P_{XY}}{P_X P_Y})$$

Proof: Let $\{(\bar{X}_i, \bar{M}_i), T_i\}_{i \in \mathbb{N}}$ be the points of a Poisson Process with intensity measure $P_X \times P_M \times \lambda_{\mathbb{R}^+}$ where P_M is the uniform distribution over $\{1, 2, \dots, L\}$.

Let's define the relationship between the random variables $\tilde{U}$ and $\{\bar{U}_i\}_{i \in \mathbb{N}}$ for an arbitrary U as follows:

$$\tilde{U} = \bar{U}_K \text{ with } K = \underset{u}{\operatorname{argmin}} \frac{Z_u}{P(u)}$$

where $\bar{U}_i$'s are i.i.d. with $\sim P(u)$ and Z_u 's are i.i.d. with $\sim \exp(1)$. Next, we choose our encoder with the input m as $\text{enc}(m) = \tilde{X}_{P_X \times \delta_m}$ where $\delta_m(m') := \mathbf{1}(\{m = m'\})$. Therefore, we have $X = \tilde{X}_{P_X \times \delta_M}$. And, we also choose our decoder as a function with input y as $\text{dec}(y) = \tilde{M}_{P_{X|Y}(\cdot|y) \times P_M}$. Therefore, we have $\hat{M} = \tilde{M}_{P_{X|Y}(\cdot|Y) \times P_M}$.

$$\implies P_e = \Pr(\{\hat{M} \neq M\}) = \Pr(\{\tilde{M}_{P_{X|Y}(\cdot|Y) \times P_M} \neq M\})$$

$$\leq \Pr(\{(\tilde{X}, \tilde{M})_{P_{X|Y}(\cdot|Y) \times P_M} \neq (X, M)\})$$

$$= E[\Pr(\{(\tilde{X}, \tilde{M})_{P_{X|Y}(\cdot|Y) \times P_M} \neq (X, M) | M, X, Y\})]$$

where the last equality is from the properties of conditional expectation. Note that the CPML is given for the quadruple of $(X, U, Y, Q_{U|Y})$. By using the CPML for the quadruple of $(M, (X, M), Y, P_{X|Y} \times P_M)$, and following the last upper bound on P_e , we get:

$$\begin{aligned} P_e &\leq E[1 - (1 + \frac{d(P_X \times \delta_M)}{d(P_{X|Y}(\cdot|Y) \times P_M)}(X, M))^{-1}] \\ &= E[1 - (1 + L \frac{P_X}{P_{X|Y}})^{-1}] = E[1 - \frac{1}{1 + L2^{-i(X;Y)}}] \end{aligned}$$

It is worth noting that we can modify our choice to decoder as a function of y as $\text{dec}(y) = \tilde{M}_{Q_{X|Y}(\cdot|y) \times P_M}$. Then, we will get:

$$P_e \leq E[1 - (1 + L \frac{P_X}{Q_{X|Y}(\cdot|Y)}(X))^{-1}]$$

This is called a "Mismatched Decoder". It is useful for the One-shot Channel Coding Problem on the Compound Channels as explained in [1].

IX. FUTURE DIRECTIONS

One of our main goals in this project is to understand the SFRL and PML with their applications given in [2] and [1]. Moreover, we aim to discover other potential applications of them. For this purpose, we studied universal channel simulation, and universal compression problems. One of our future directions in this project is studying some problems and bringing them new perspectives with SFRL and PML techniques. We are planning to study some communication theory problems as [1] and [2] mostly did and we are also planning to study some security problems such as Secret Key Agreement and some guessing problems such as the one discussed in [5]. The other future direction is finding some kind of universal version for PML as we did for SFRL.

A. Secret Key Generation from Joint Randomness

Secret Key Generation Problem from Common Randomness is introduced in [6]. Note that this work is way before than [2] and [1]. We are thinking that SFRL might be a useful tool for this problem since it allows us to create a random variable Z that does not contain any information about X , therefore it can be used as a Secret Key, and it is still quite informative about Y . We will investigate ways to use SFRL for the Secret Key Generation Problem and compare them with the methods used in [6].

B. Guessing and Its Applications to Sequential Coding

In his paper [5], Arikan sets up a guessing problem and presents some asymptotical results. He also gives some applications of it to sequential coding. This problem is worth studying and it is possible to acquire some one-shot results by using functional representation techniques for the price of getting a looser bound on the expected number of guesses. In this case, it might have some potential applications to sequential coding.

Moreover, we can consider a guessing problem of guessing between 2 distributions p and q . In the interest of making it relative to the SFRL, we can consider these distributions as $p = p_{XY}$ and $q = p_X p_Y$. Here, let's assume there is 2 party, namely Alice and Bob. Bob is trying to guess between distributions p_{XY} and $p_X p_Y$ by observing a realization of Y^n and nR bits sent by Alice who observes a realization of X^n . In this setup, it is shown that:

$$P(\text{error}) \leq 2^{-ne(R)} \text{ with } e(R) = \inf_{U-X-Y \text{ and } I(U;X) \leq R} I(U;Y)$$

As we can see, this is an asymptotical results. By studying this problem and trying to rederive it with the SFRL, we are hoping to get a bound of $2^{-ne'(R)}$ on the error probability where $e'(R) = \inf_{\{U-X-Y \text{ and } f(I(U;X)) \leq R\}} I(U;Y)$ with $f(x) := x + \log(x+1) + c$ for some constant c .

C. Universal version of PML

In this project, we presented a universal version of the Strong Functional Representation Lemma, namely USFRL, which adapts to a wide range of source distributions without requiring specific prior knowledge. Building on it, we are planning to explore the development of a universal version of the Poisson Matching Lemma. This potential universal version will be useful for the cases where we can not apply the original PML in some scenarios where the distribution of the source is unknown.

In conclusion, we studied SFRL and PML given in [2] and [1] with some of their applications. We also searched for other potential applications of them and presented our results in this report. We are planning to continue studying other potential applications of these theorems as well as universal versions of them to make them more robust. Particularly, we are planning to study the problems discussed in IX and, hopefully, get some meaningful, and potentially one-shot, results for these problems utilizing SFRL, USFRL, and PML.

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